

Glossary

Air resistance A force that pushes against an object that is moving through the air, slowing it down. This is also called drag.

Alloy A mixture of two or more metals.

Alternating current (AC) An electrical current that changes its direction of flow many times every second to allow the efficient use of electricity.

Altitude The height of an object relative to sea or ground level.

Ammonia A colorless gas made up of nitrogen and hydrogen. It is often used to make fertilizers that aid plant growth.

Anesthetic A pain-relieving drug or gas given to a patient before a surgical operation. It can be “local,” which affects just one part of the body, or “general,” so the person is made temporarily unconscious before a major operation.

Ancestor A relative—usually one more remote than a grandparent—from whom a human is descended. An ancestor can also refer to animals and plants.

Anode The positively charged electrode in a cell, such as a battery.

Antenna This device is used to transmit or receive radio or television signals. It is also known as an aerial.

Antibiotic A medicine that kills—or limits the spread of—bacterial infections in the human body or other animals.

Antiseptic A substance that is put on cuts to prevent an infection.

Archaeology The study of things that show how people lived before there were written records.

Astronomy The study of objects in space, including planets, stars, and galaxies.

Atom The smallest part of a chemical element, made up of particles called protons, neutrons, and electrons.

Bacteria (singular: bacterium) These are tiny, single-celled organisms. Some bacteria cause disease, while others help and protect the human body.

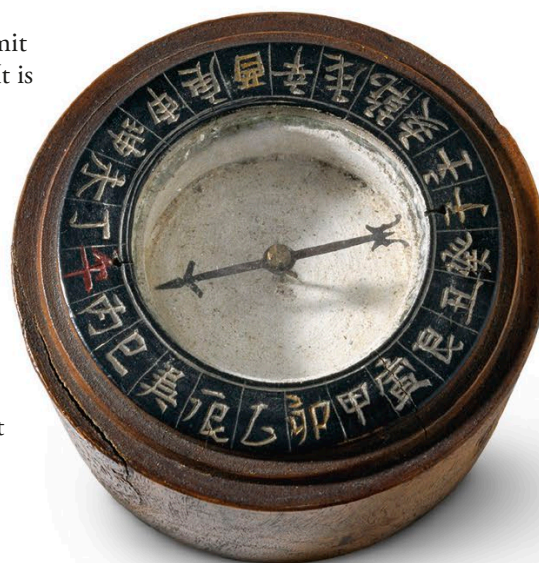
Battery A portable container of chemicals that stores and supplies electricity to a range of devices, from toys to cars.

Bellows A device that expands to draw in air through a valve, then contracts to force the air through a tube.

Bipedal A human, animal, or something else—for example, a robot—that uses two legs for walking.

Bitumen This sticky, black substance made from tar or gasoline is used mainly to surface roads.

Bronze A yellow-brown metal alloy made up mostly of copper and tin. It is a popular choice of material for sculptures because it is strong, hard-wearing, and rust-resistant.



▲ MID-19TH CENTURY CHINESE OCEAN COMPASS

Bronze Age A historical period, between the Stone Age and Iron Age, that is defined by the use of bronze as the most important material for making weapons and tools. It began around 3000 BCE in the Middle East and spread around the world over about 2,500 years.

Carbon An important chemical element that comes in a variety of forms, from coal to diamonds. Carbon can also be combined with itself and other elements to produce millions of different compounds, from plastics to DNA.

Cathode The negatively charged electrode in a cell, such as a battery.

Celestial A way of describing an object or event that occurs in the sky or outer space, such as a comet or an eclipse.

Charcoal This gray-black porous solid, made by heating wood in the absence of oxygen, is mostly carbon. It has several uses, such as burning to generate heat for warmth or to cook. It can also be used as a material for drawing.

Circuit A complete and closed path that electricity flows around. All electric and electronic things have circuits inside them.

Combustion A chemical reaction in which a fuel, such as wood or coal, burns with oxygen from the air to release heat energy.

Conductor A substance that lets heat or electricity flow easily through it.

Contamination The process of polluting or poisoning—for example, an oil spill polluting the ocean, or harmful bacteria poisoning the human body.

Coolant A substance, usually a liquid, used to reduce the temperature of something, such as a car engine or factory machine.